

A Report Card for the Night Sky



One score for how good a night of observing was. It blends image quality, sky coverage, and time well spent into a single number an AI telescope scheduler can improve. Built on 105,889 DES exposures.

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THE IDEA · one number for a whole night

How good was the night? We ask three simple questions and turn the answers into one score the AI scheduler can chase.

① Clean, steady images

Was the sky clear, the stars sharp, the night dark? Each image is scored by how much real signal it actually caught.

② The whole sky, evenly

Credit goes to filling thin, unfinished patches, not re-shooting the easy ones. Even coverage is what the science needs.

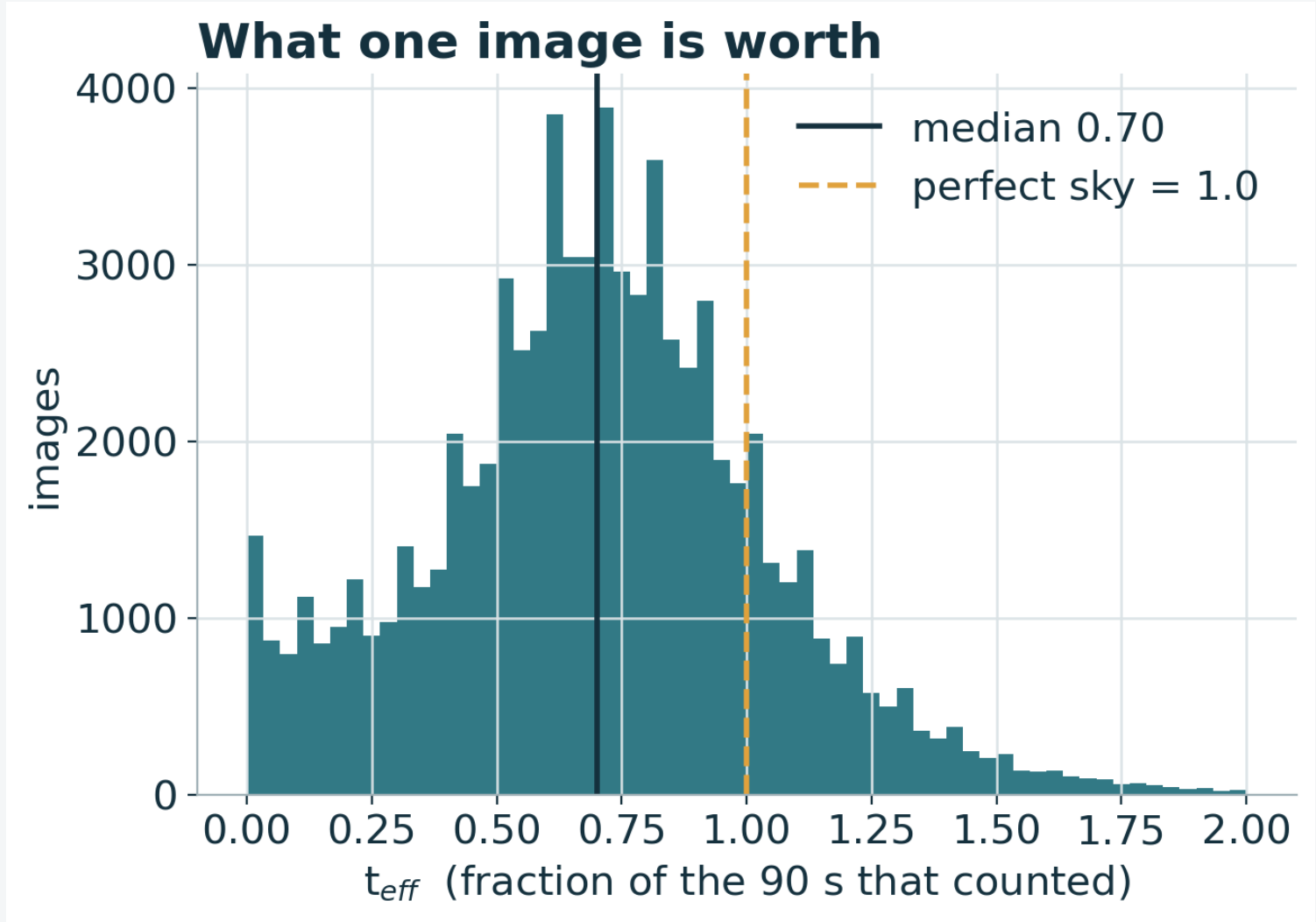
③ Little wasted time

How much of the night's hours became real science, instead of readout, slews, and overhead.

Together they make one score, Q, that climbs when a night banks more good science per hour, spread evenly across the sky..

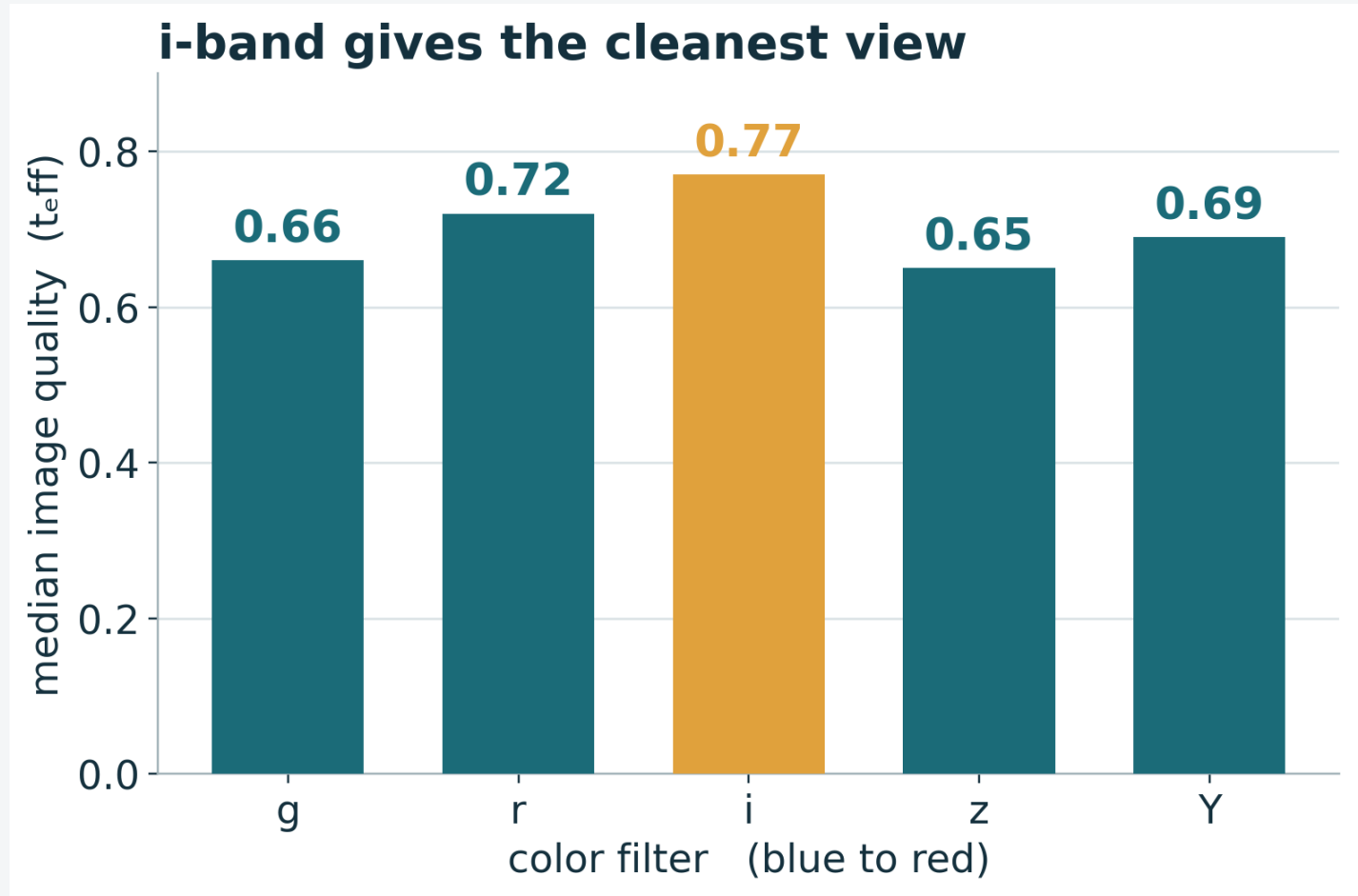
PART 1 · ONE IMAGE

How good was each image?



Every exposure gets a score, t_{eff} , the share of a full 90 second image that actually counted after cloud, blur, and bright sky. Median 0.70.

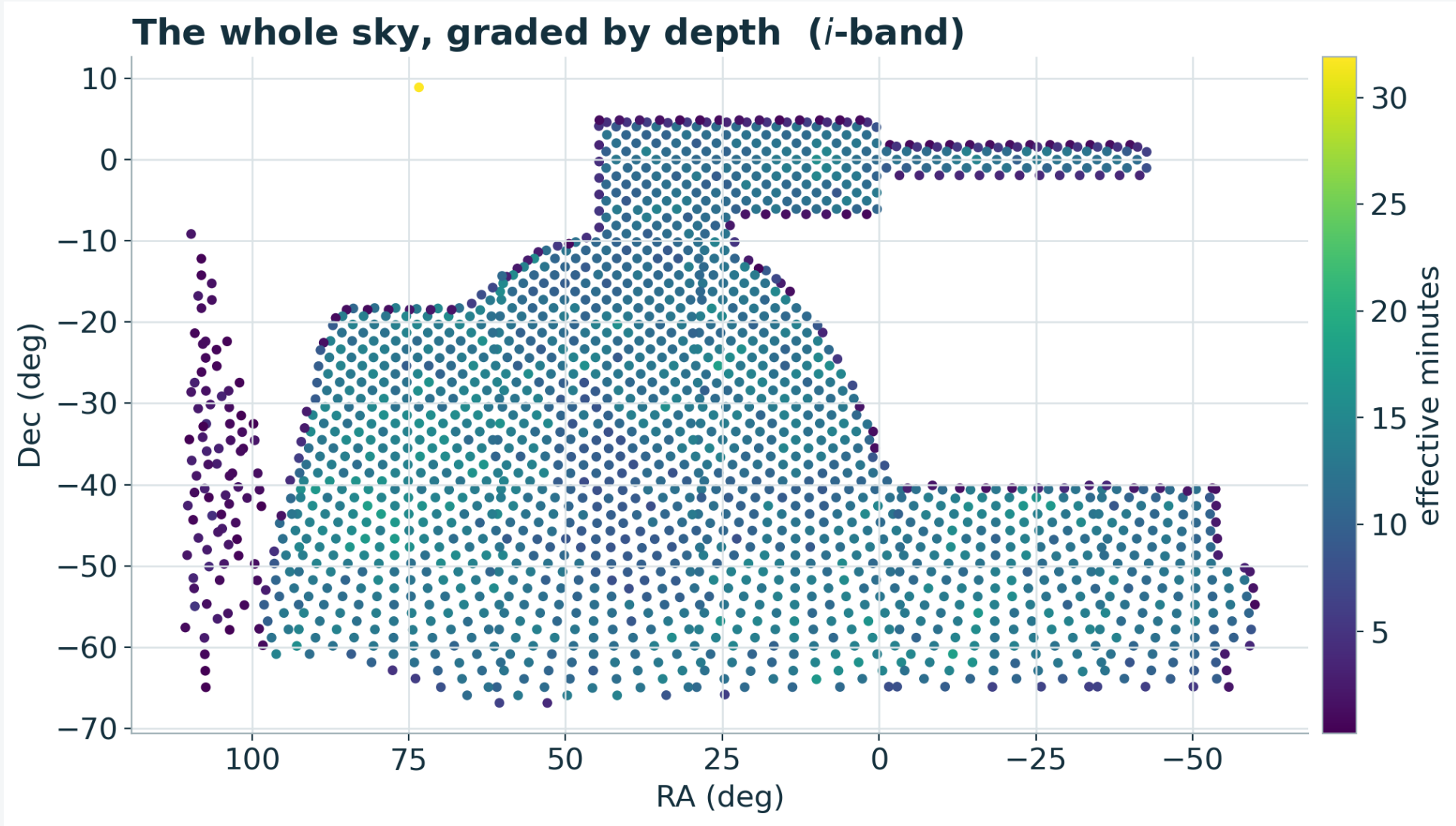
Which filter does best?



The same score, split by the survey's five color filters. i-band scores highest, which is why the depth map is shown in i-band.

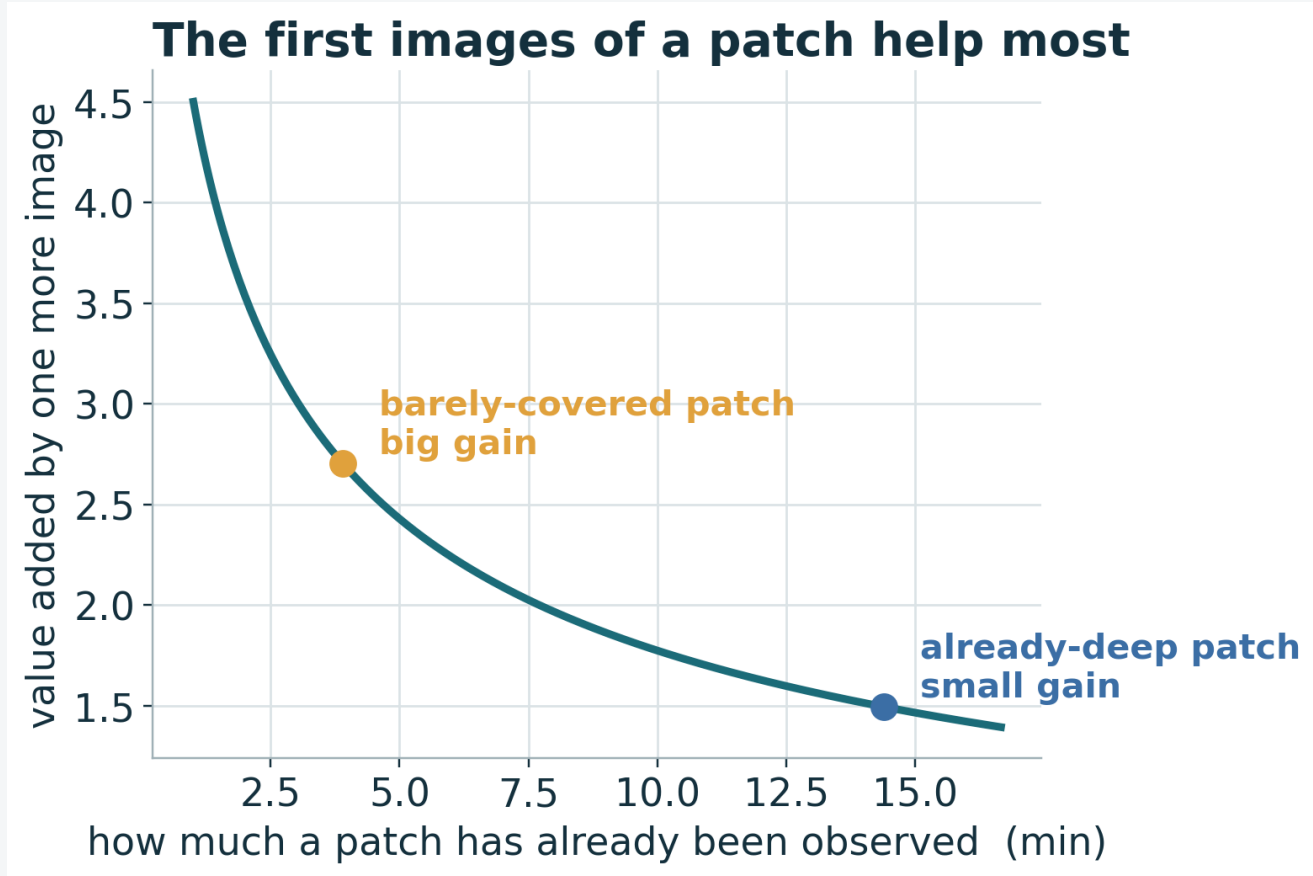
PART 2 · THE WHOLE SKY

Building the whole sky



Add up each image's worth everywhere the telescope pointed and you get a depth map, how much good exposure every patch has banked. Median about 12 minutes.

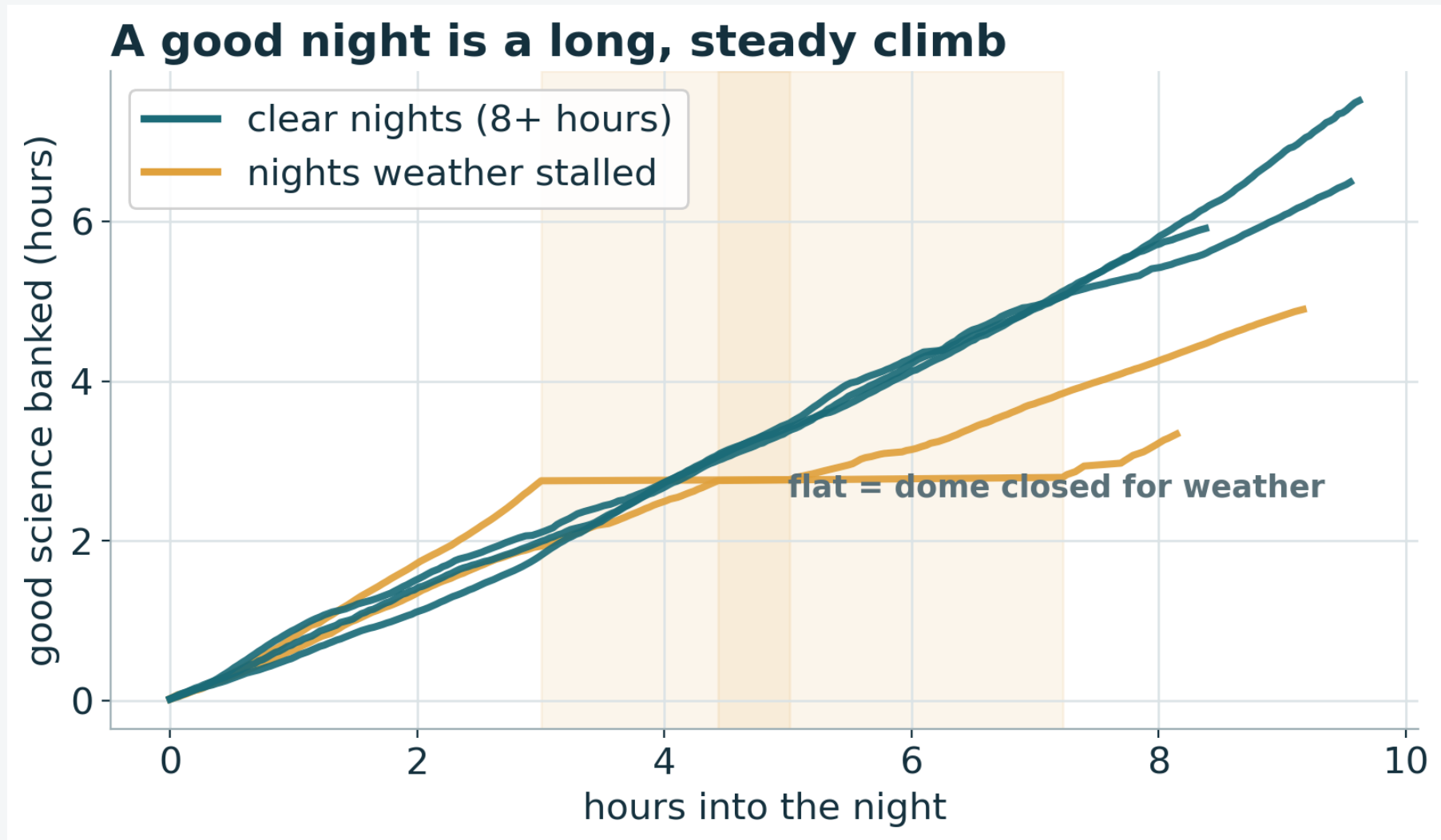
Why even coverage wins



The first images of a patch help the most. Piling more onto an already deep patch helps less, so the score rewards finishing thin sky. That sky is worth about 1.8 times more.

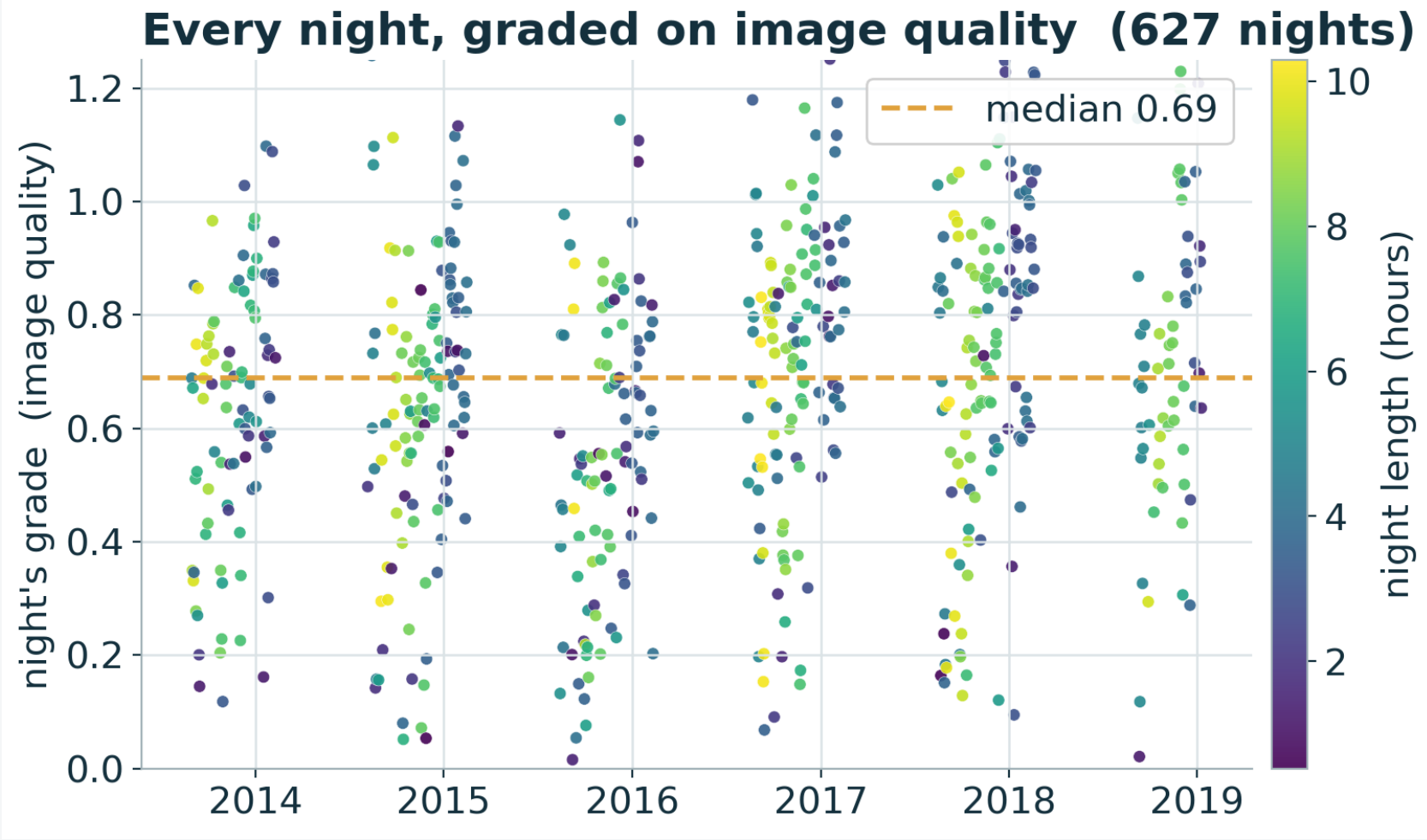
PART 3 · A WHOLE SCHEDULE

What a good night looks like



Each line is one night, banking science hour by hour. Clear nights climb steadily for 8+ hours; the flat stretches are weather, not the telescope.

Every night, graded



Graded on image quality, median 0.69. Night length barely matters (color), so half nights are as fair as full ones.